```
PS C:\Windows\system32> nslookup -type=NS iitb.ac.in
Server:  UnKnown
Address:  10.28.124.244

Non-authoritative answer:
iitb.ac.in      nameserver = dns1.iitb.ac.in
iitb.ac.in      nameserver = dns2.iitb.ac.in
iitb.ac.in      nameserver = dns3.iitb.ac.in
PS C:\Windows\system32>
```

Q4

```
PS C:\Windows\system32> nslookup -type=A www.iitb.ac.in 10.28.124.244
Server:  UnKnown
Address:  10.28.124.244

Non-authoritative answer:
Name:    www.iitb.ac.in
Address:  103.21.124.133
```

Q1-Q2-Q3

——Questions

1. Run nslookup www.iitb.ac.in (the Indian Institute of Technology Bombay). What IPv4 address does it return?

The IPv4 address returned by the DNS lookup is 103.21.124.133.

2. What is the IP address of the DNS server that gave you that answer?

The IP address of the local DNS server is 10.28.124.244.

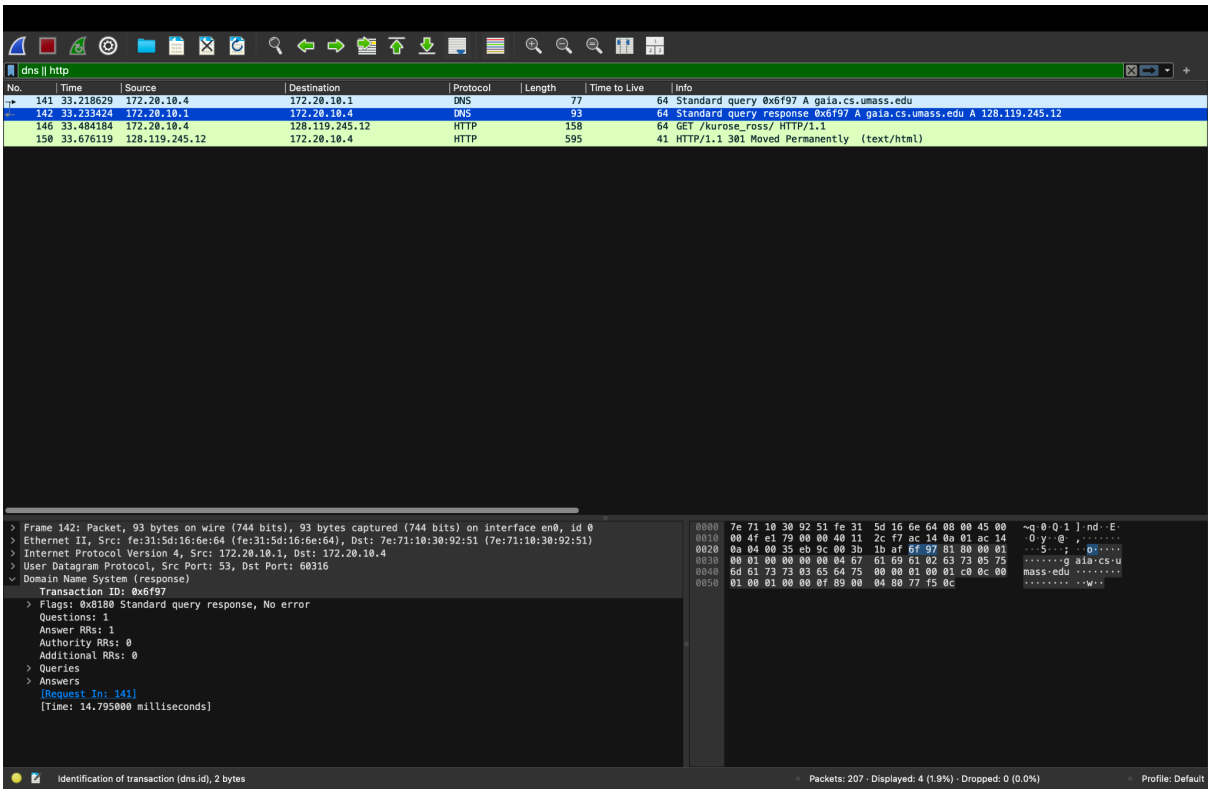
3. Was the answer authoritative or non-authoritative? What does that tell you about where it came from?

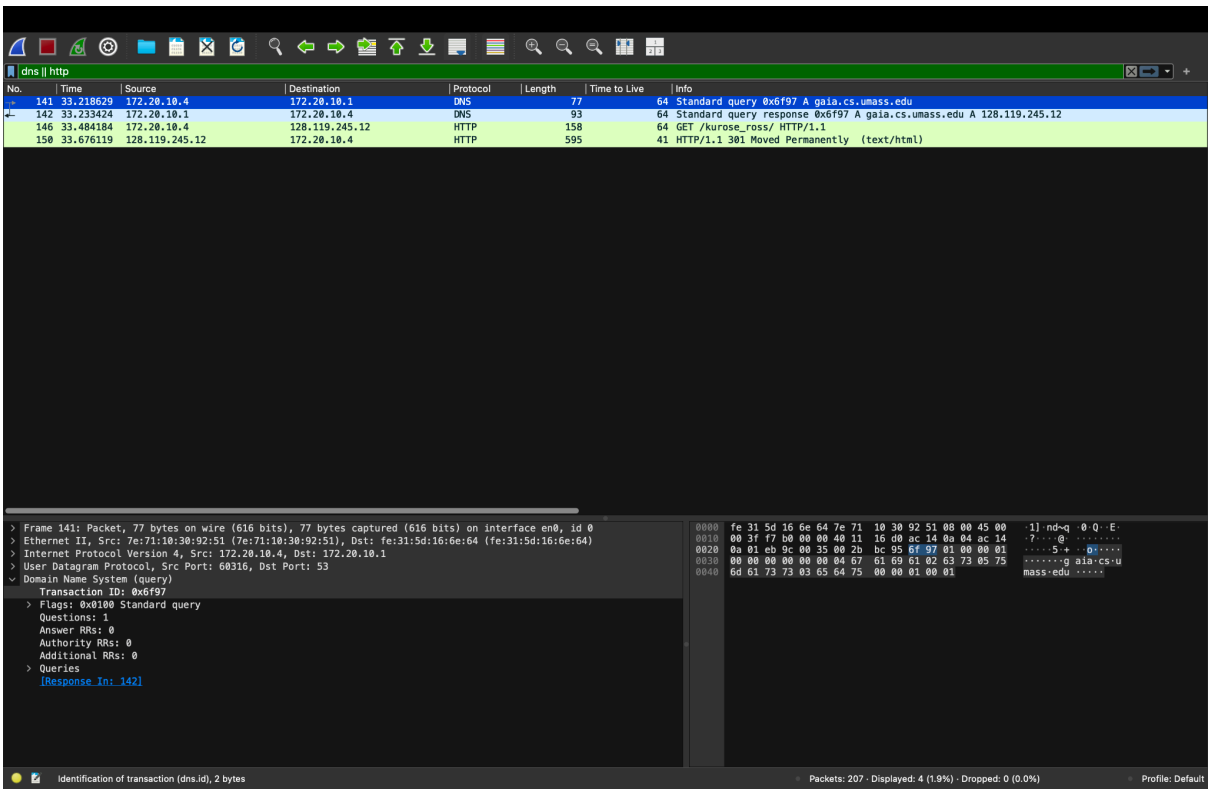
The answer was non-authoritative. This means the response came from the local DNS resolver rather than directly from an authoritative DNS server for the domain. The resolver may have obtained the result from its cache.

4. Use nslookup -type=NS iitb.ac.in to find the authoritative name servers for the iitb.ac.in domain. What is the name of the first one listed? Which command would you run to get that server's IP address?

The first authoritative name server listed is dns1.iitb.ac.in. To find its IP address, I used the following command: `nslookup dns1.iitb.ac.in`

The command returned the IPv4 address 103.21.125.129.





5. Find the first DNS query for gaia.cs.umass.edu. What is its packet number? Is it carried over UDP or TCP?

Packet number: 141

It is carried over UDP.

6. Find the matching DNS response. What is its packet number? UDP or TCP? (Tip: Wireshark links the pair. Look for “Response In” in the query’s details.)
Packet Number: 142
Carried over UDP

7. What is the destination port of the query? What is the source port of the response?
Destination Port: 53
Source Port: 60316

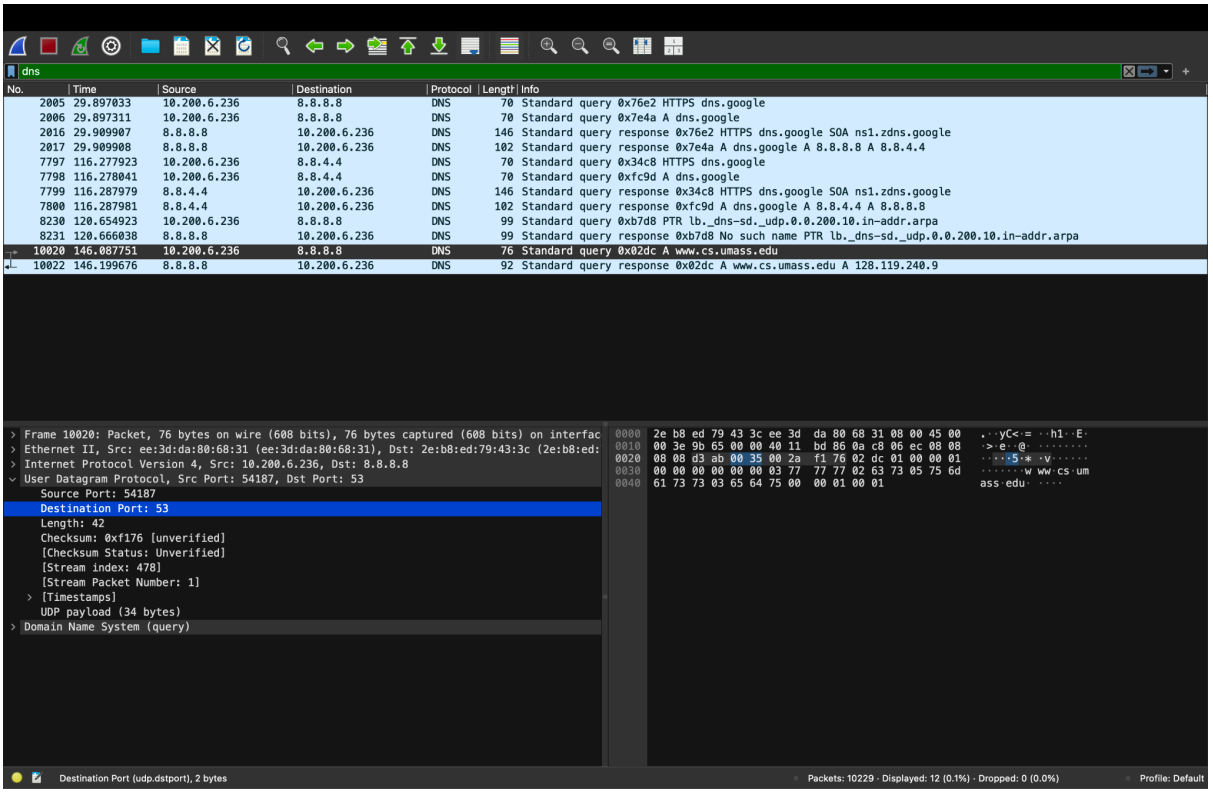
8. To which IP address was the query sent? Compare it with your local DNS server from question 2.
To: 172.20.10.1
It is the same one with the local DNS server from question 2.

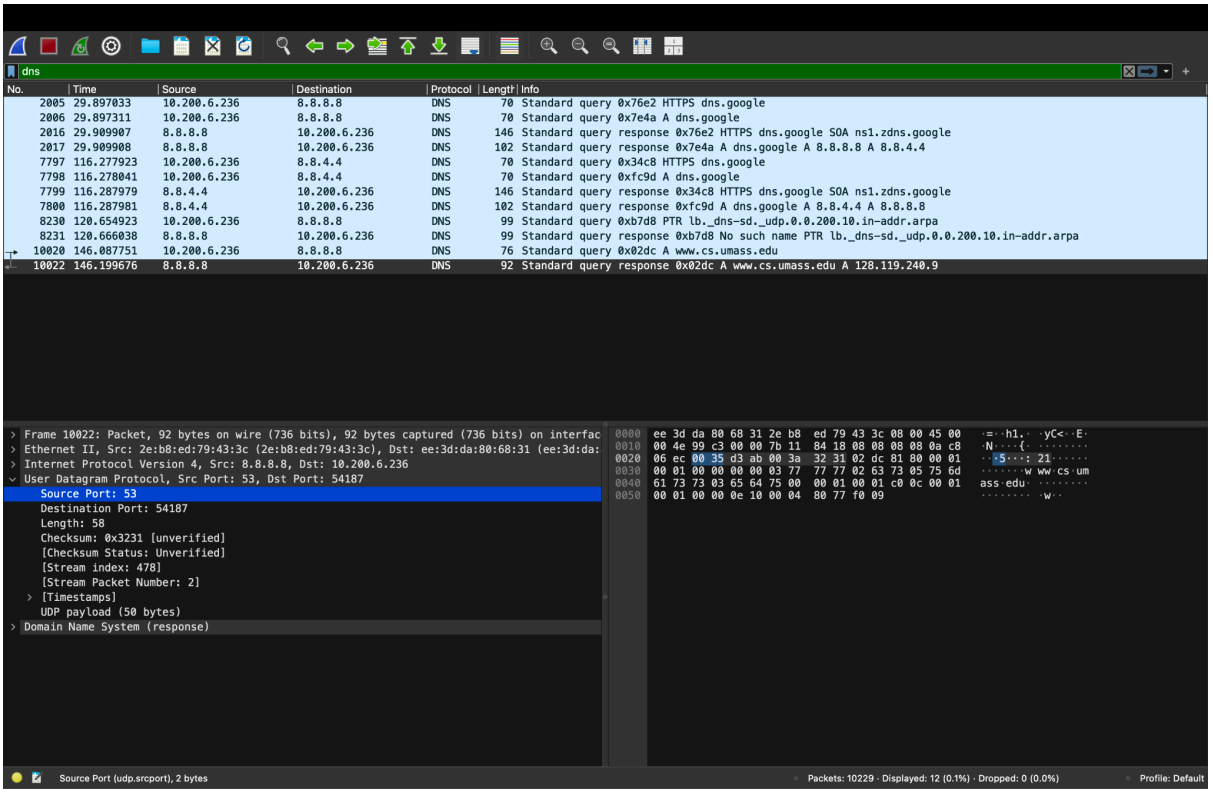
9. In the query message, how many Questions and how many Answer RRs are there?
Questions: 1
Answer RRs: 0

10. In the response message, how many Questions and how many Answer RRs are there?
Question: 1
Answers RRs: 1

---

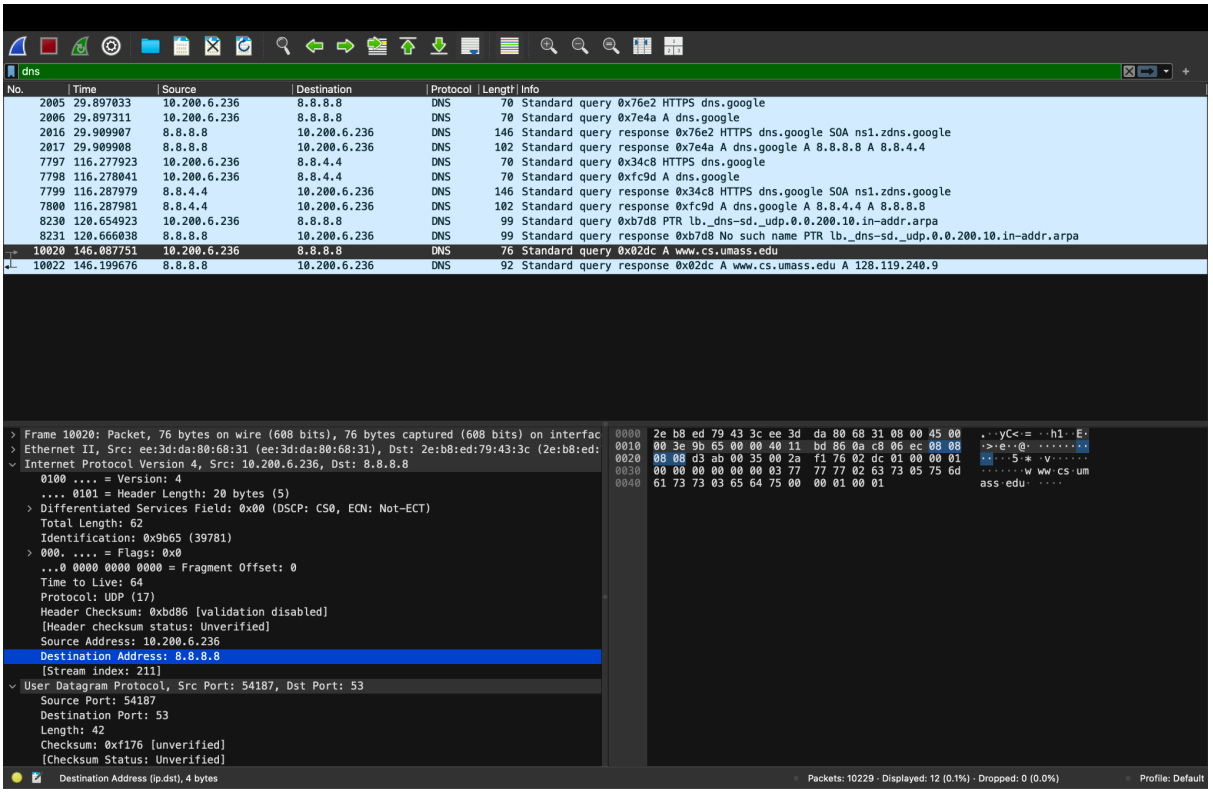
11. What is the destination port of the query? What is the source port of the response?

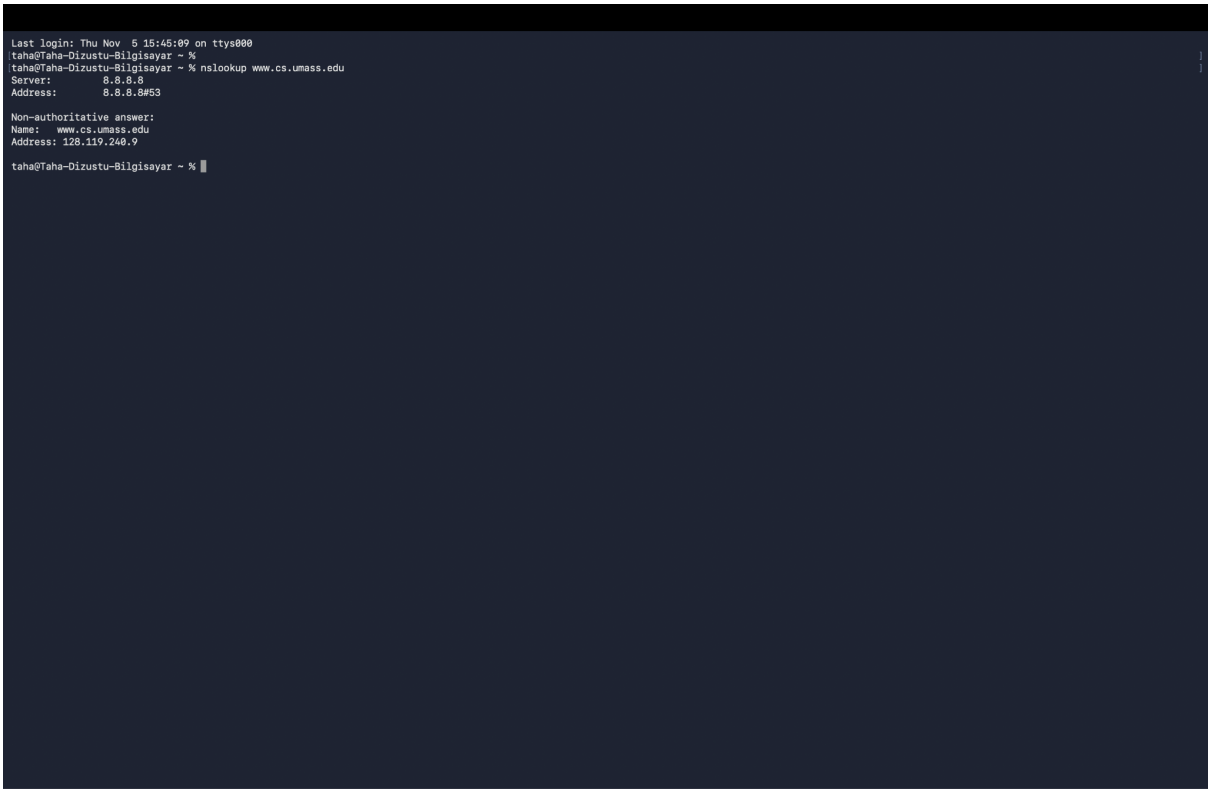




Destination Port: 53
Source Port: 53

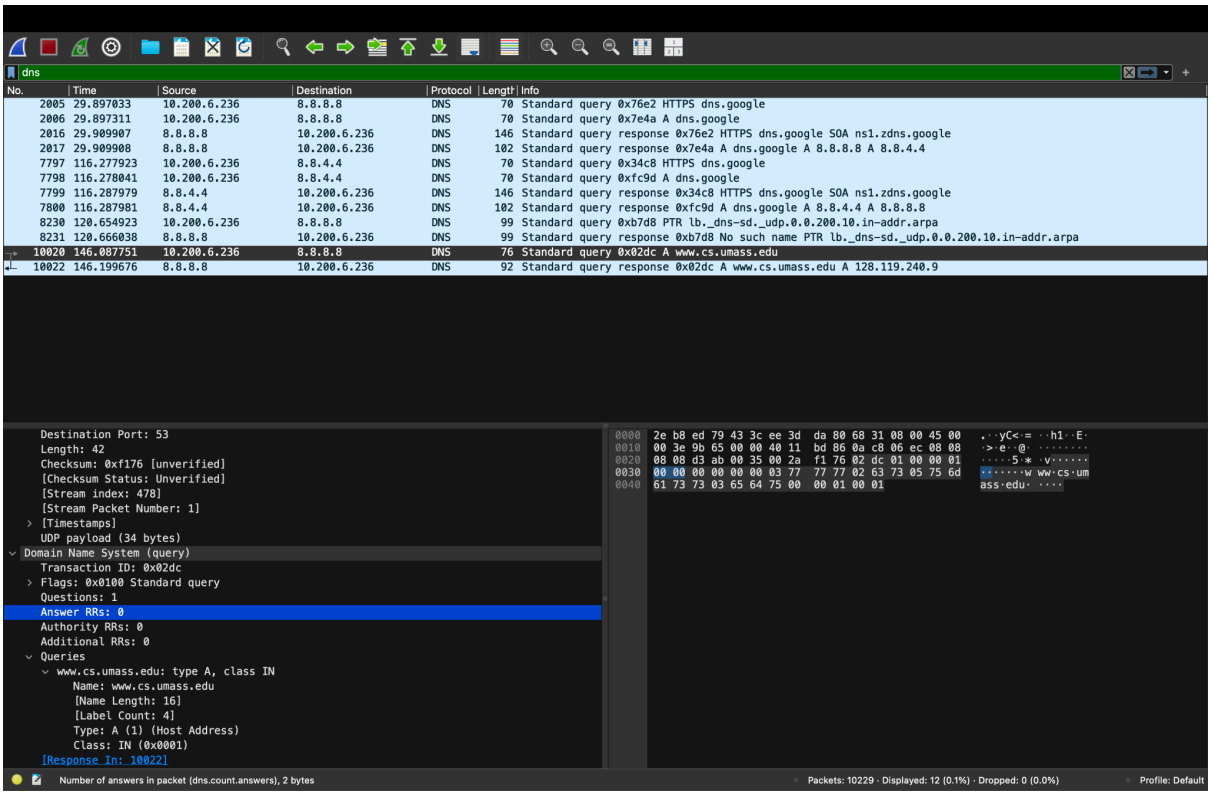
12. To which IP address was the query sent? Is it your default local DNS server?

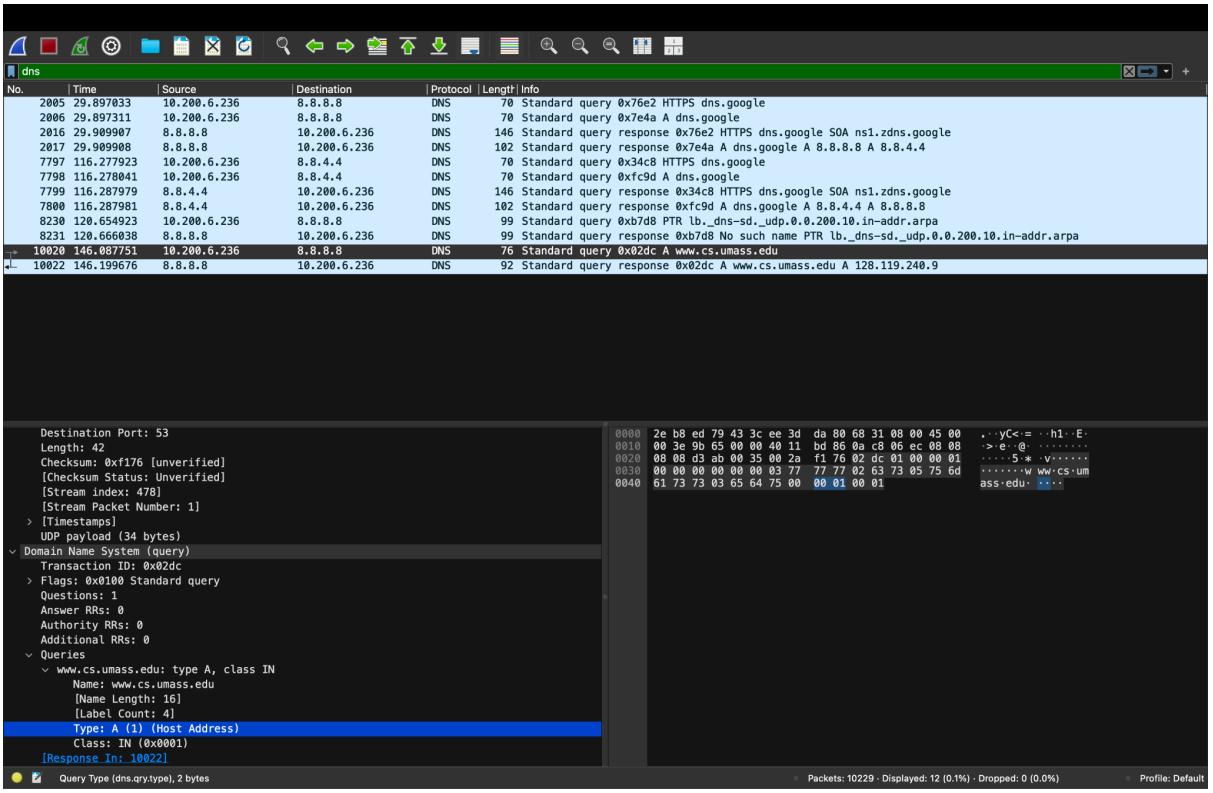




The query was sent to 8.8.8.8. Yes, this is my default local DNS server, because it matches the Address shown at the top of the nslookup output.

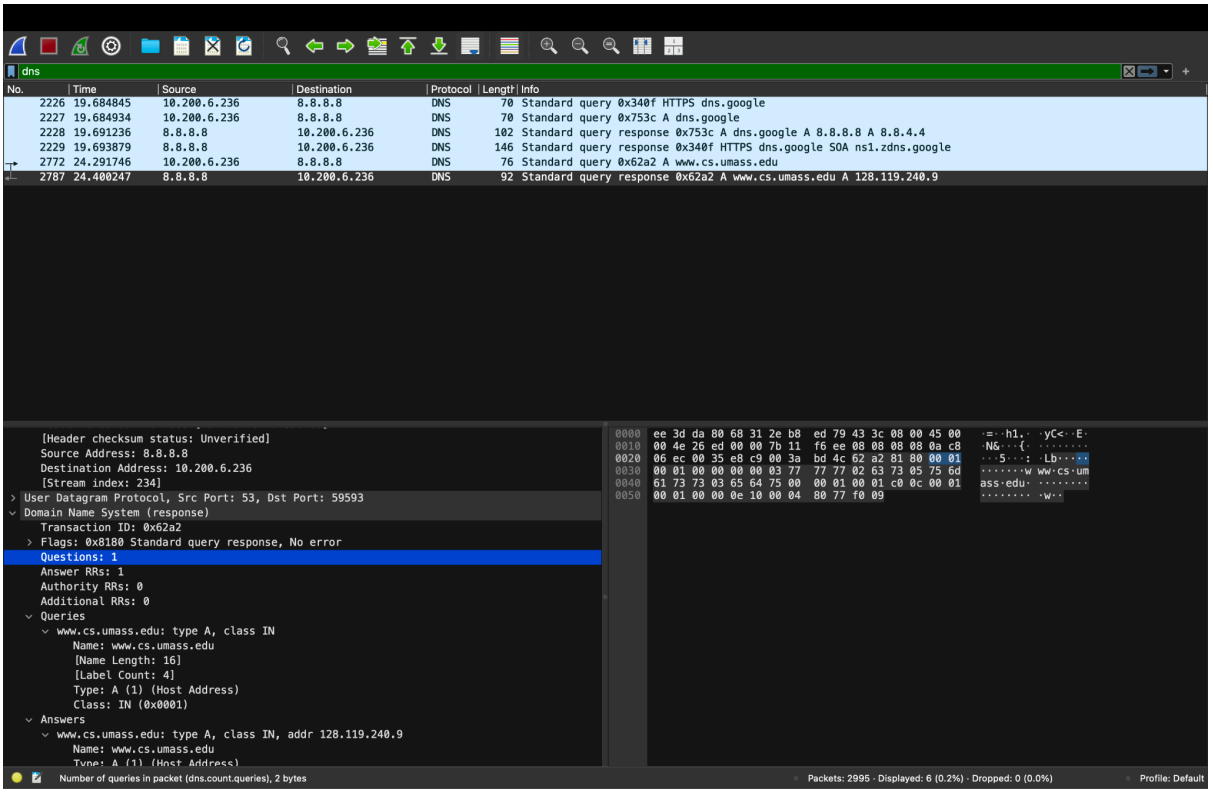
13. What Type is the query? Does the query contain any answers?

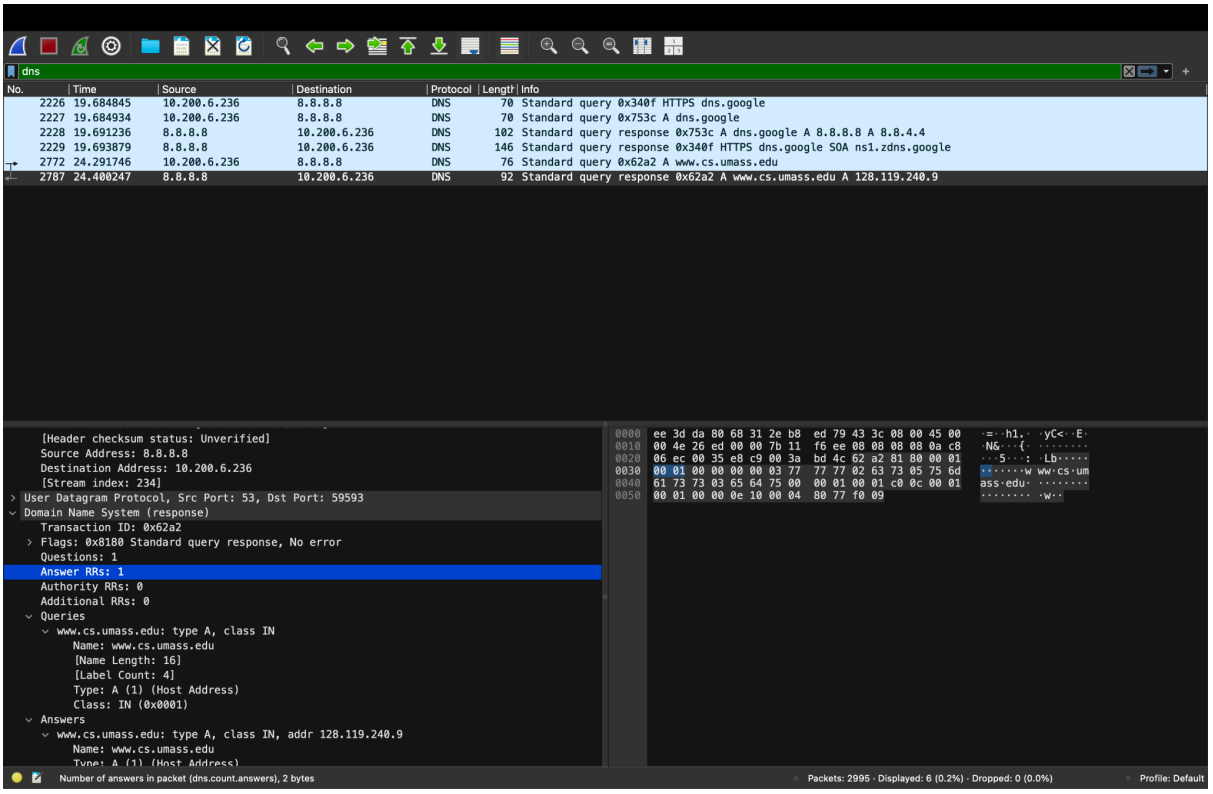


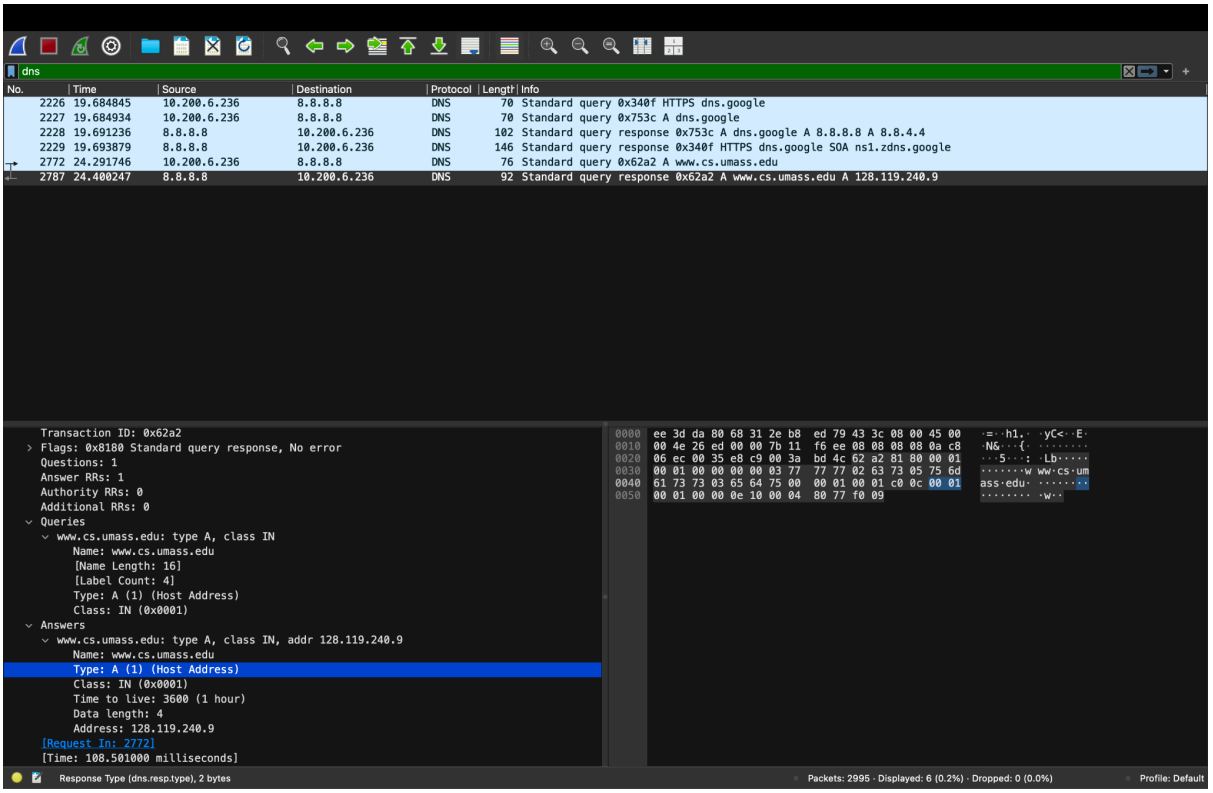


The query is of Type A (IPv4 host address). No, the query does not contain any answers (Answer RRs: 0). It only contains one question.

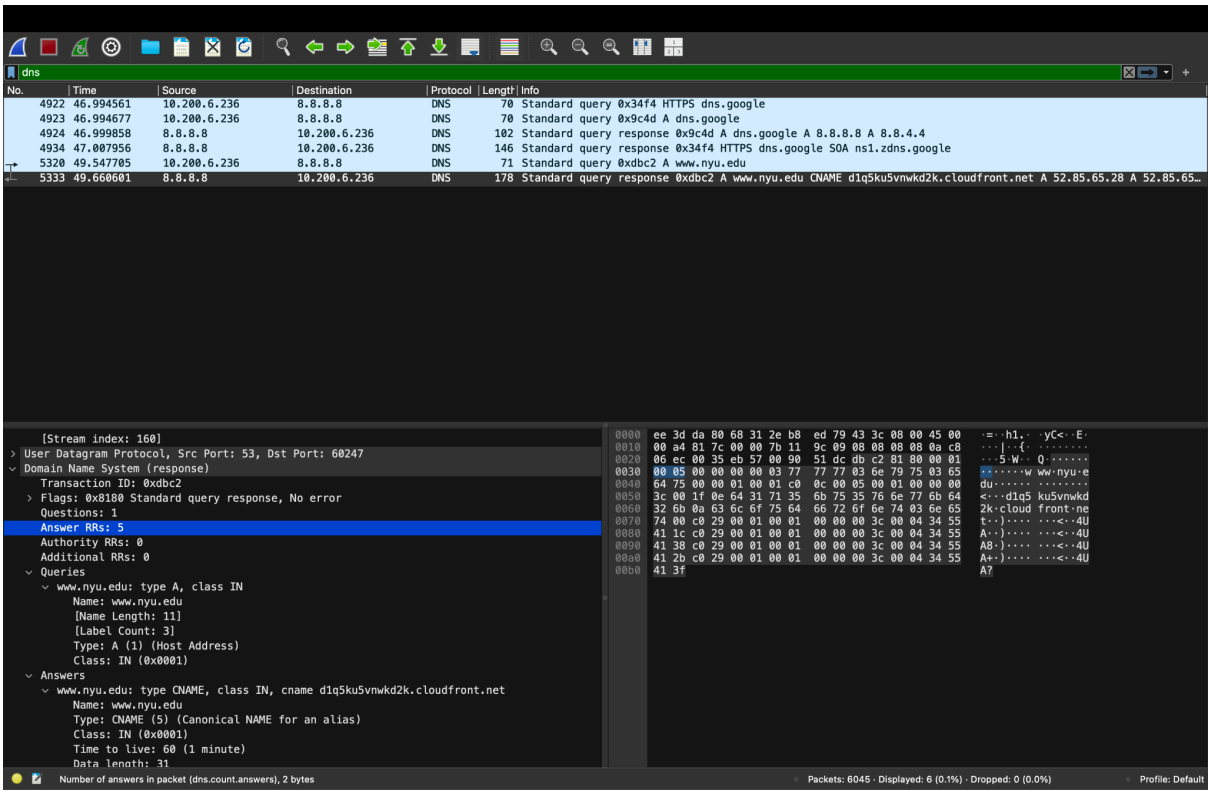
14. In the response, how many Questions and how many Answers are there? If there is more than one answer, what record types are they? Now capture nslookup www.nyu.edu the same way.
Its response contains CNAME records. How many answers does it have, and what does a CNAME record tell you about the name www.nyu.edu?

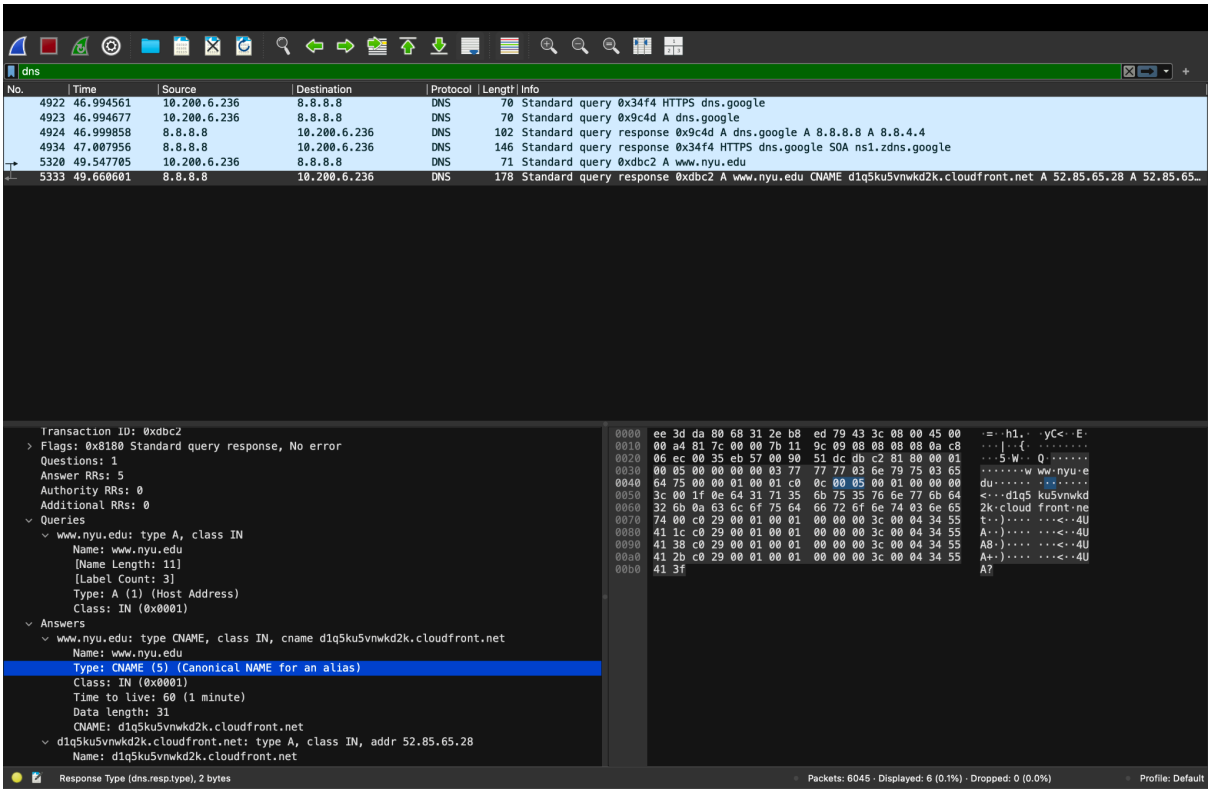


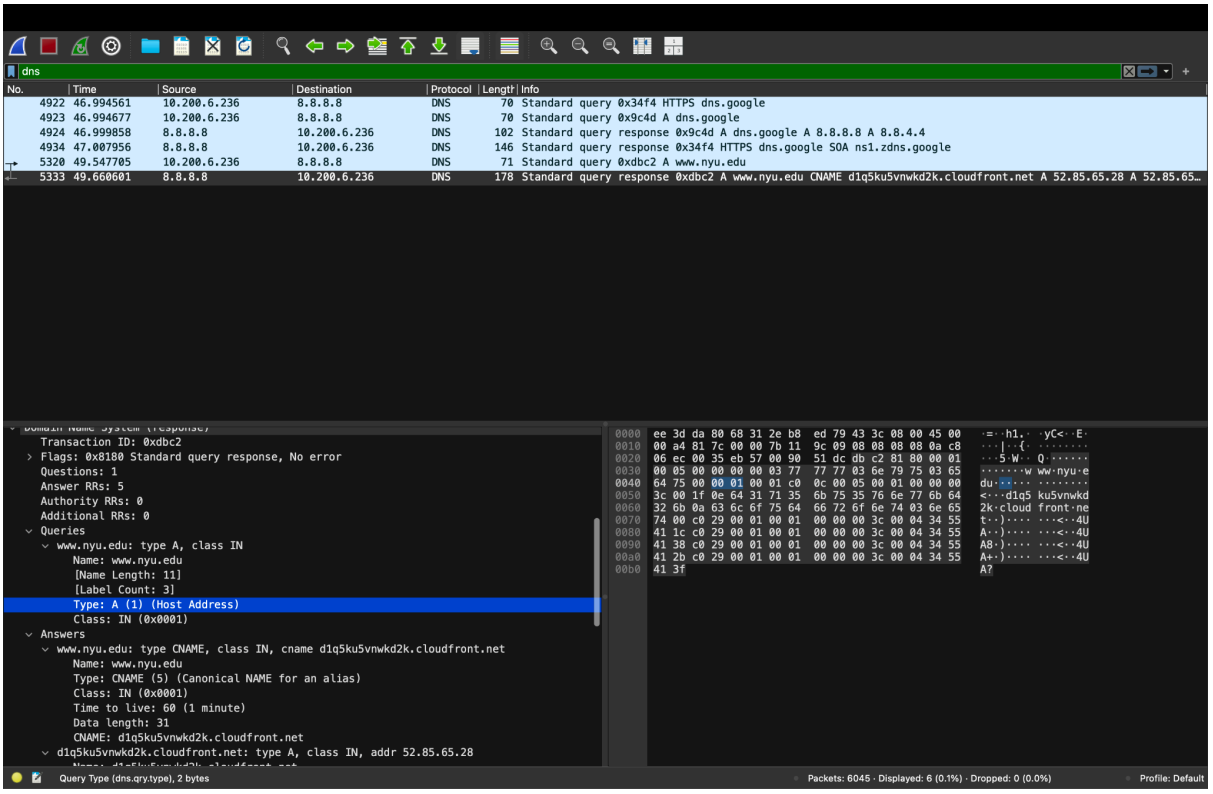




For [www.cs.umass.edu](): Questions: 1, Answers: 1. The single answer is a Type A record (IPv4 address).







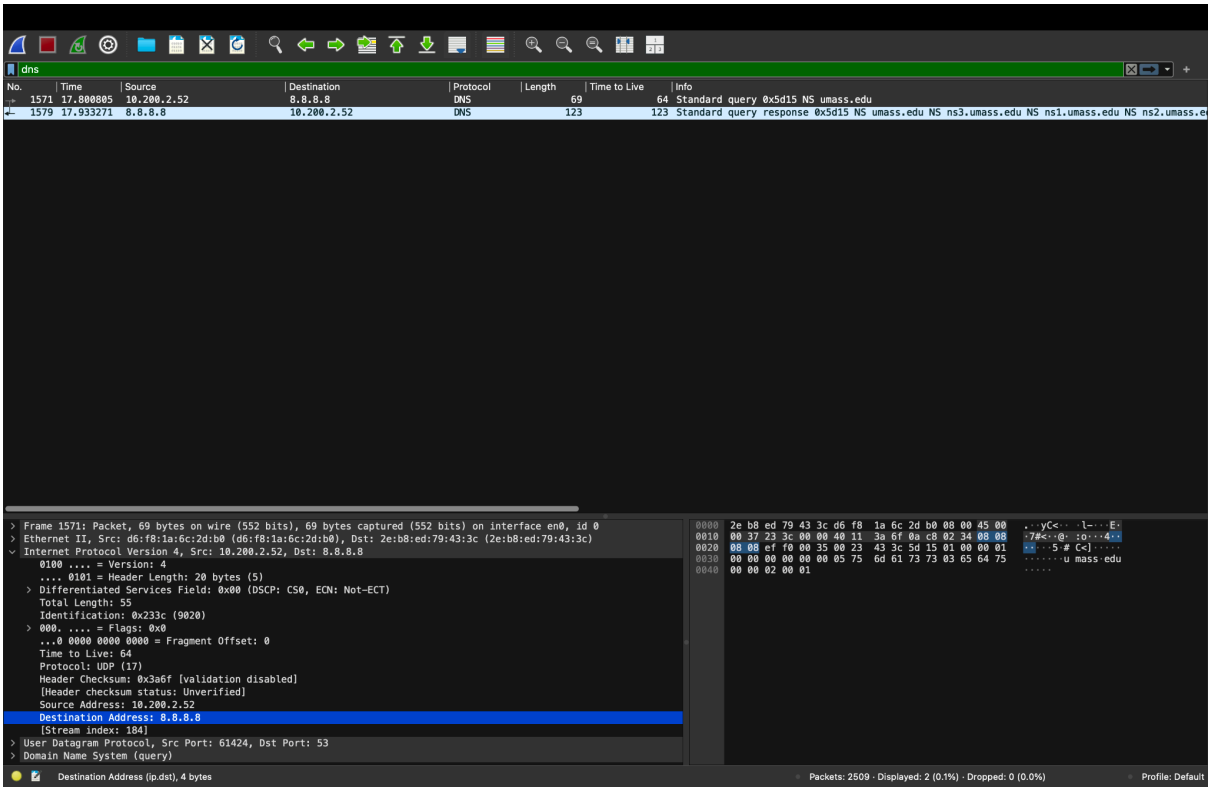
[www.cs.umass.edu](): The response contains 1 question and 1 answer. The single answer is a Type A record (IPv4 address).

[www.nyu.edu](): The response contains 1 question and 5 answers: 1 CNAME record and 4 A records.

A CNAME record tells us that `www.nyu.edu` is only an alias, not the real (canonical) name of the server. It points to the canonical name `d1q5ku5vnwkd2k.cloudfront.net`, and the IP addresses (52.85.65.28, 52.85.65.56, 52.85.65.43, 52.85.65.63) are given in the A records of that canonical name.

---

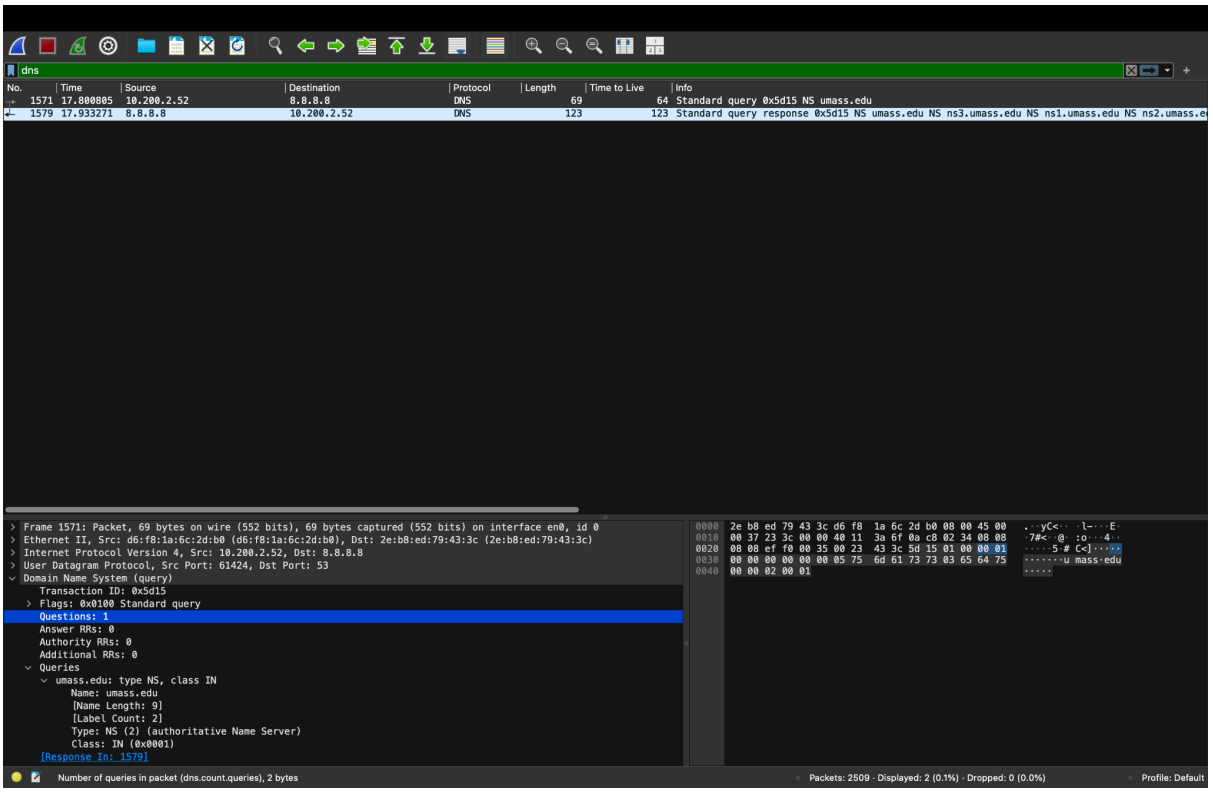
15. To which IP address was the query sent? Is it your default local DNS server?
IP Adress: 8.8.8.8
Yes it is.

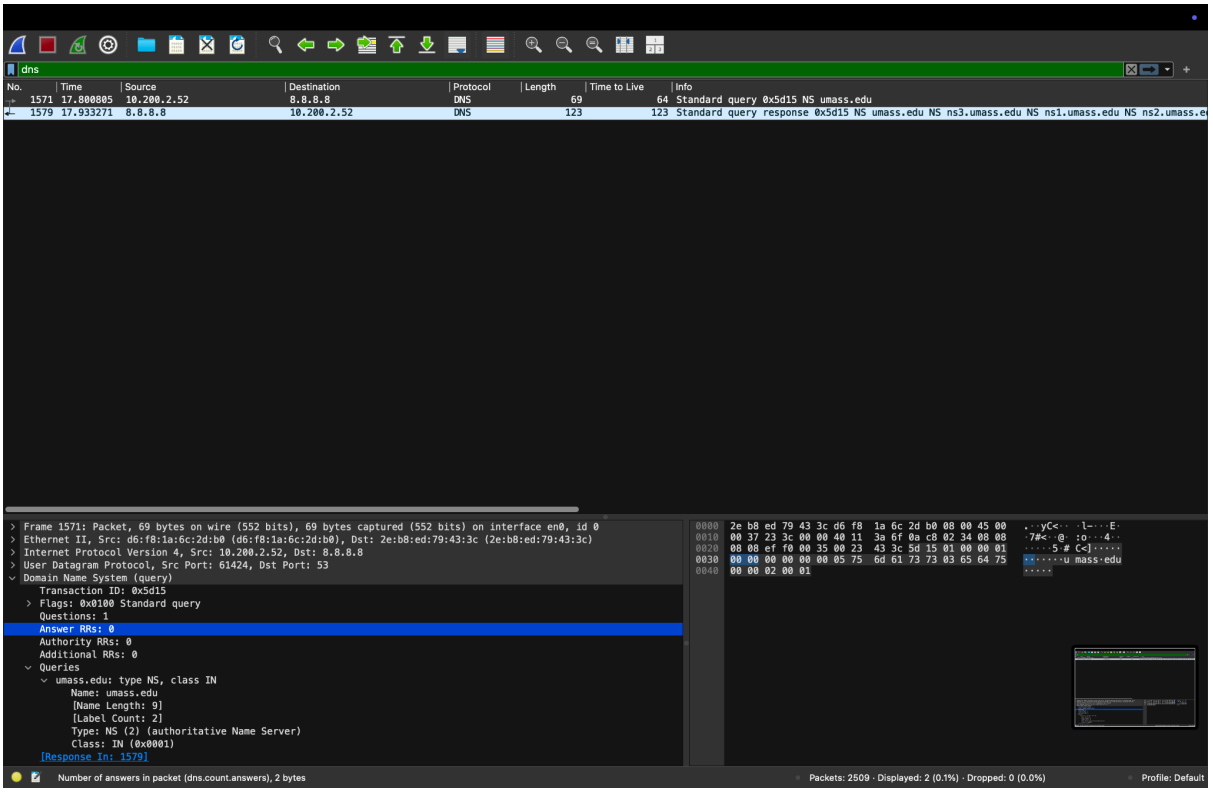


16. How many questions does the query contain? Does it contain any answers?

Question: 1

Answer RRs: 0 (No it does not.)

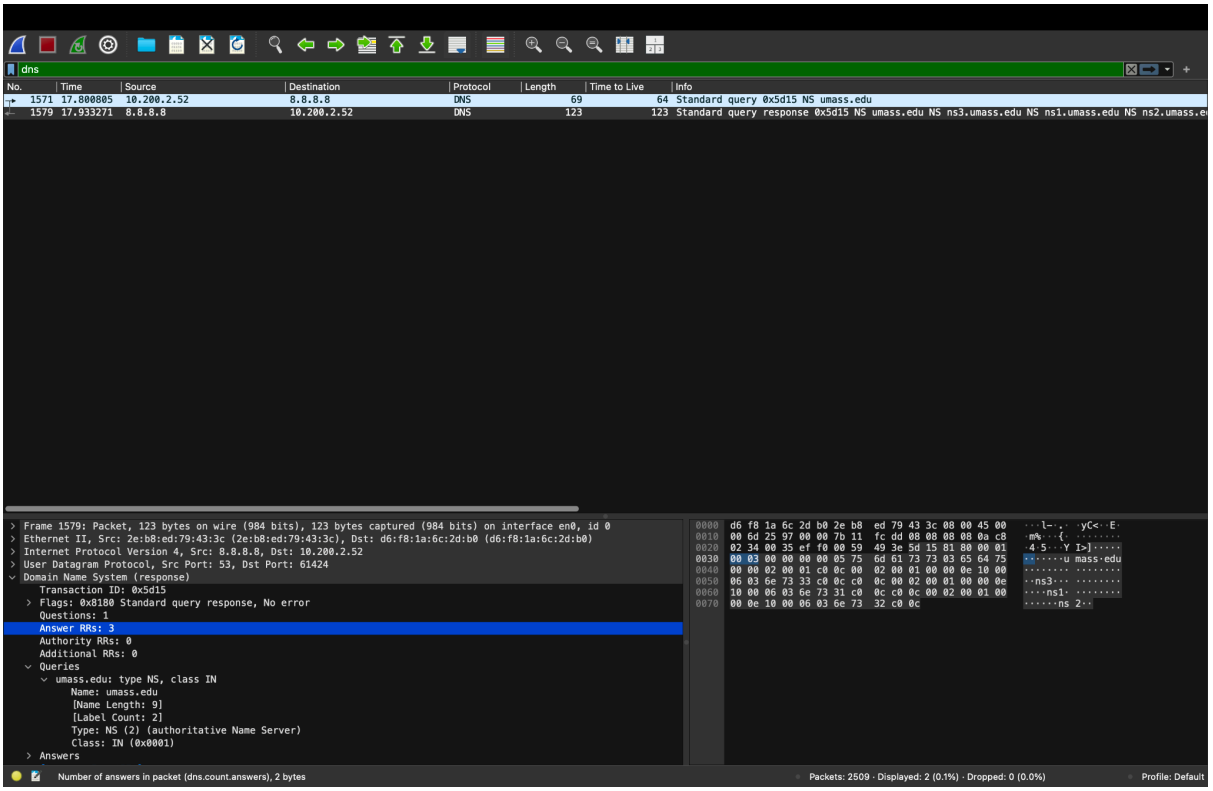


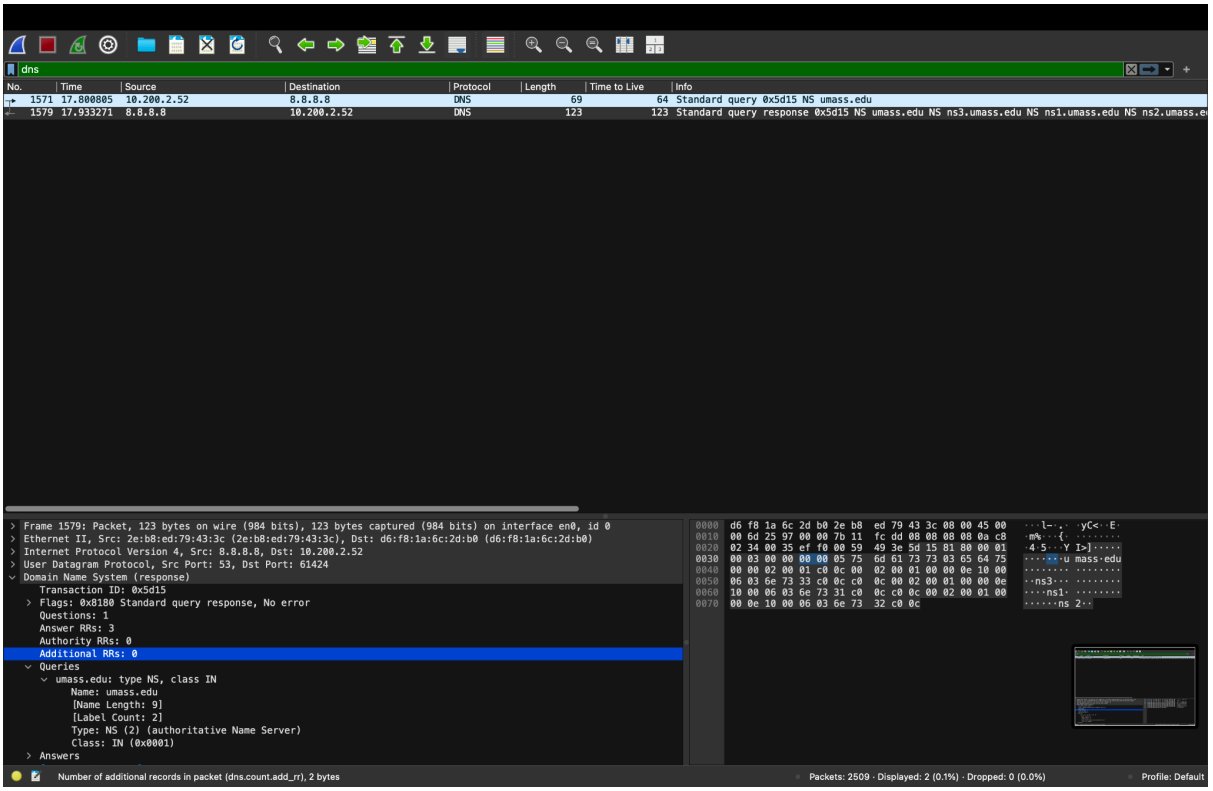


17. Look at the response of type NS. How many answers does it have, and what information do they contain? How many Additional RRs are returned, and what do they contain?

Answer: 3 questions: ns3.umass.edu, ns1.umass.edu, ns2.umass.edu

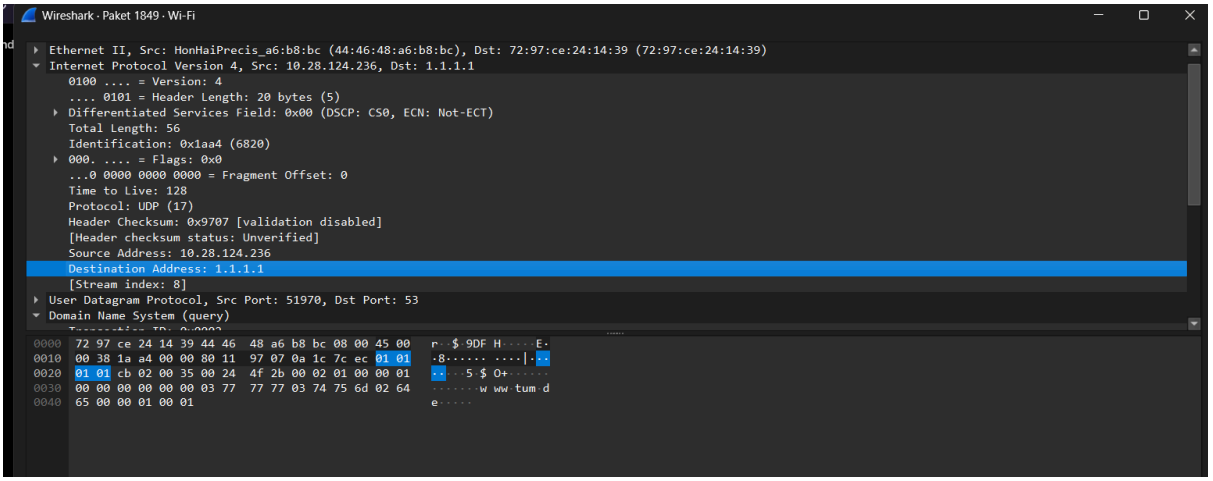
Additional RRs: 0, there are no RRs, so they cannot contain any information.





B1. Ask a different DNS server. You can name the server to ask as a second argument:
nslookup www.tum.de 1.1.1.1
nslookup www.tum.de 8.8.8.8
Who operates 1.1.1.1 and 8.8.8.8? Do the answers match the one from your local DNS server
(nslookup www.tum.de)? Capture one of these queries in Wireshark. What destination IP do you
see now?

1.1.1.1 is operated by Cloudflare, and 8.8.8.8 is operated by Google. All three DNS servers returned the same IPv4 address, 129.187.255.109, for www.tum.de. In Wireshark, the destination IP of the Cloudflare DNS query was 1.1.1.1.



B2. Watch the TTL count down. Every DNS answer carries a TTL (time to live): how many more seconds it may be kept in a cache. Look up www.lmu.de twice, about 30 seconds apart, and write down the TTL each time:
• macOS / Linux: dig www.lmu.de. In the ANSWER SECTION, the number after the name is the TTL.
• Windows (PowerShell): Resolve-DnsName www.lmu.de. Read the TTL column.
Check one of the lookups in Wireshark as well: in the response, open Domain Name System → Answers → the record, and find Time to live.

What happened to the TTL between the two lookups, and why? What does this tell you aboutcaching in the local DNS server?
Note: if the TTL jumps up instead of going down, your network may use several DNS servers, each with its own cache. Another possibility is that the cached record expired between your two lookups. If you see two answers (a CNAME and an A record), use the TTL of the A record.

The TTL of the A record decreased from 20594 to 20565 seconds, a difference of 29 seconds. This happens because DNS records are cached and their remaining TTL decreases over time. It shows that the DNS resolver is using a cached record.

```
PS C:\Windows\system32> Resolve-DnsName www.lmu.de

Name                           Type   TTL   Section    NameHost
----                           ----   ---   -------    --------
www.lmu.de                     CNAME  81    Answer     webfelrz-cluster0.verwaltung.uni-muenchen.de

Name       : webfelrz-cluster0.verwaltung.uni-muenchen.de
QueryType  : AAAA
TTL        : 20594
Section    : Answer
IP6Address : 2001:4ca0:4f06:1:141:84:44:56


Name       : webfelrz-cluster0.verwaltung.uni-muenchen.de
QueryType  : A
TTL        : 20594
Section    : Answer
IP4Address : 141.84.44.56



PS C:\Windows\system32> Resolve-DnsName www.lmu.de

Name                           Type   TTL   Section    NameHost
----                           ----   ---   -------    --------
www.lmu.de                     CNAME  52    Answer     webfelrz-cluster0.verwaltung.uni-muenchen.de

Name       : webfelrz-cluster0.verwaltung.uni-muenchen.de
QueryType  : AAAA
TTL        : 20565
Section    : Answer
IP6Address : 2001:4ca0:4f06:1:141:84:44:56

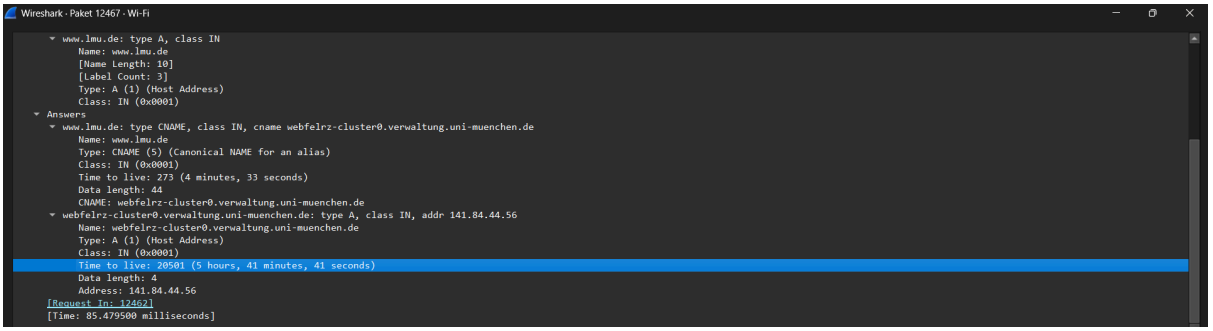

Name       : webfelrz-cluster0.verwaltung.uni-muenchen.de
QueryType  : A
TTL        : 20565
Section    : Answer
IP4Address : 141.84.44.56



PS C:\Windows\system32> Resolve-DnsName www.lmu.de

Name                           Type   TTL   Section    NameHost
----                           ----   ---   -------    --------
www.lmu.de                     CNAME  273   Answer     webfelrz-cluster0.verwaltung.uni-muenchen.de

Name       : webfelrz-cluster0.verwaltung.uni-muenchen.de
QueryType  : AAAA
TTL        : 20501
Section    : Answer
IP6Address : 2001:4ca0:4f06:1:141:84:44:56
```

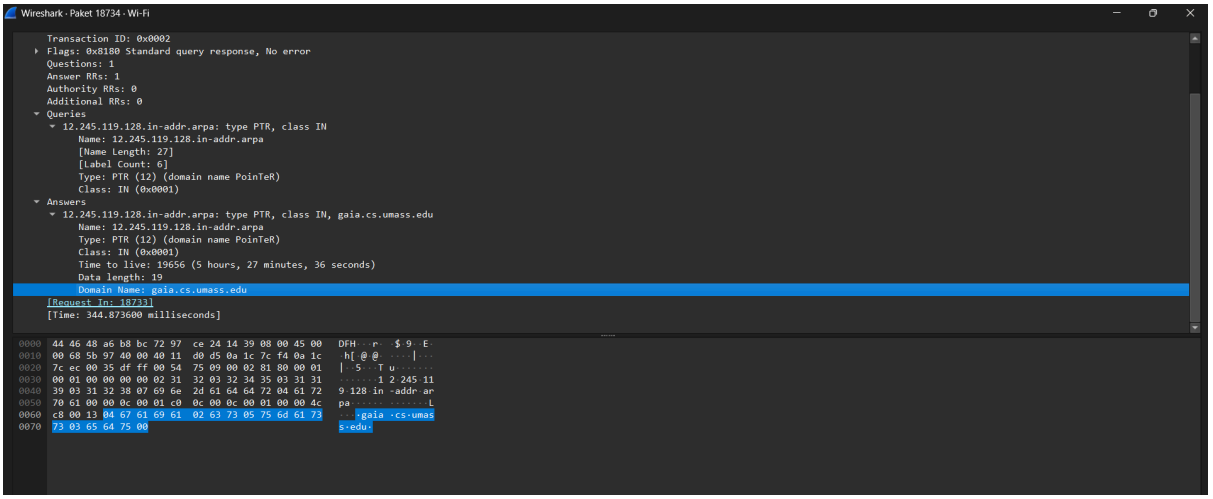


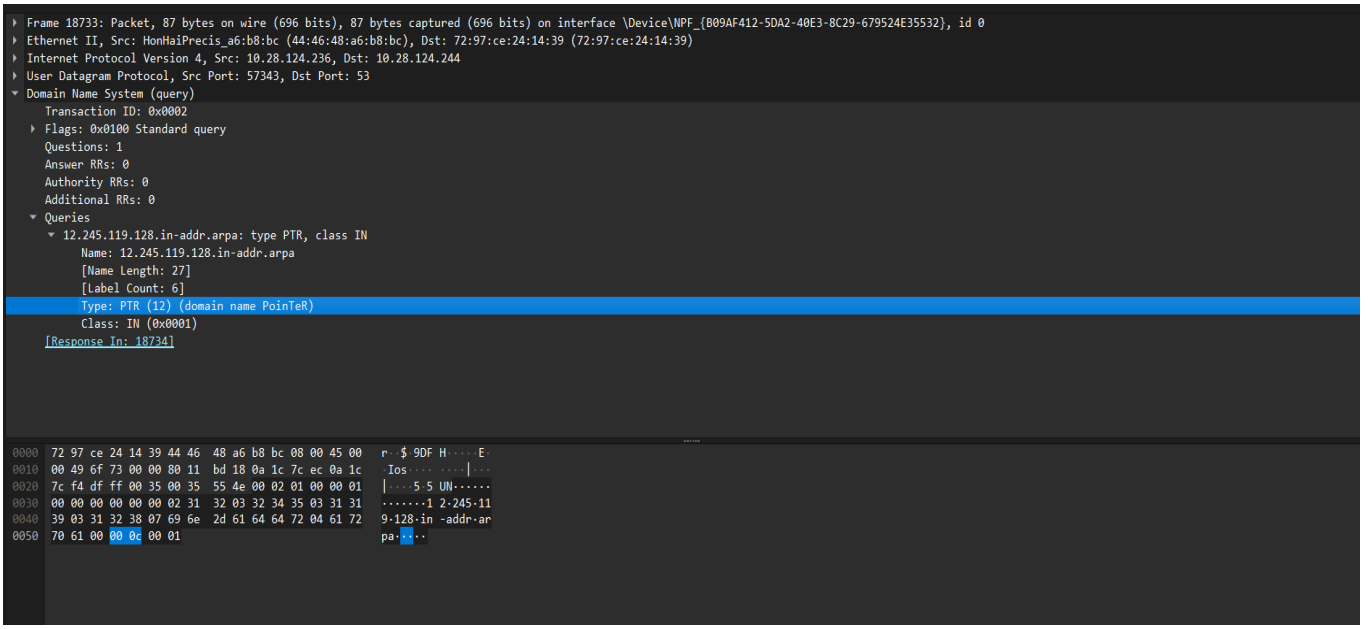
B3. Reverse lookup. Capture in Wireshark while you run nslookup 128.119.245.12. Which hostname comes back? In Wireshark, what record Type does a reverse lookup query use, and what name is actually asked for? (Hint: look for in-addr.arpa.)

```
PS C:\Windows\system32> nslookup 128.119.245.12
Server:  UnKnown
Address:  10.28.124.244

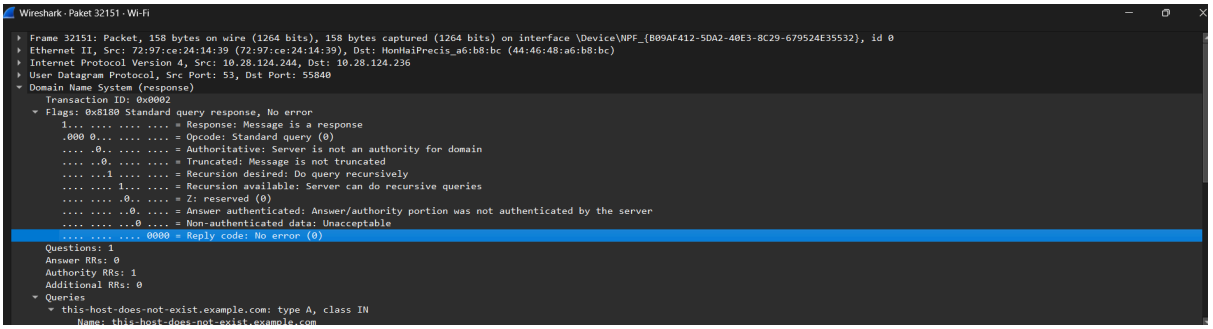
Name:    gaia.cs.umass.edu
Address:  128.119.245.12
```

The reverse lookup of 128.119.245.12 returned gaia.cs.umass.edu. Wireshark showed that the query used a PTR record and requested 12.245.119.128.in-addr.arpa.





B4. A name that doesn’t exist. Run nslookup this-host-does-not-exist.example.com. In Wireshark, find the response. What does the Reply code field in the flags say?



The DNS response showed the reply code "No error (0)" with zero answer records. This means the DNS server did not return an IP address for the requested hostname, but it did not report an NXDOMAIN error.

B5. Reflection (2–3 sentences). The DNS messages you captured in this lab were not encrypted. What could someone who sees your network traffic (for example, the Wi-Fi operator) learn about you from your DNS queries?

Someone monitoring my network could see which websites I visit through my DNS queries. They could get an idea of my interests and online habits, but they wouldn't be able to see exactly what I do on those websites.